

ABDELKHALK ESSAID

Data Science & Machine Learning Engineer — Full-Stack Developer

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PROFILE

Master's student in Data Science & Information Systems Security with a software-engineering background and hands-on experience across full-stack development, backend engineering, and applied machine learning. Built an intelligent Industry 4.0 supervision platform combining 3D simulation, IoT telemetry, and ML-based anomaly detection as a final-year project. Seeking a junior Data Science / Machine Learning or Full-Stack Developer role to apply this cross-disciplinary foundation on production systems.

INTERNSHIP EXPERIENCE

Full-Stack & ML Developer — Virtual FabLab, Final-Year Project (ENSA / FPBM)

Feb – Jun 2026

Béni Mellal

- Designed and built a full-stack Industry 4.0 platform for supervising a virtual fabrication lab: machine reservations, admin dashboards, interactive 3D simulation, and simulated IoT telemetry.
- Implemented an ML module for predictive maintenance using Isolation Forest anomaly detection, plus a generative-AI chatbot, served through a FastAPI backend over MQTT telemetry.

[Next.js](#) · [React Three Fiber](#) · [TypeScript](#) · [Tailwind CSS](#) · [FastAPI](#) · [Python](#) · [MQTT](#) · [scikit-learn](#)

Software Development Intern — OCP Group, Laverie Daoui, Khouribga

Mar – May 2023

- Built a desktop and mobile application (Java, Android Studio, JavaFX, Spring Boot) to manage equipment visit workflows and monitor industrial equipment status for field agents and administrators.

Web Development Intern — Factory Gear, Tanger

Aug – Sep 2022

- Designed and developed a responsive static website showcasing the company's maintenance, automation, and robotics services, improving its online presence.

EDUCATION

Master's, Data Science & Information Systems Security

Sep 2024 – Present

Faculté Polydisciplinaire, Université Sultan Moulay Slimane — Béni Mellal

Licence (Excellence), Data Science & Information Systems Security

Sep 2023 – Jun 2024

Faculté Polydisciplinaire, Université Sultan Moulay Slimane — Béni Mellal

DUT, Software Engineering

Sep 2021 – Jun 2023

École Supérieure de Technologie (EST), Université Sultan Moulay Slimane — Béni Mellal

TECHNICAL SKILLS

Programming	Python, Java, JavaScript/TypeScript, SQL, PHP, C++
Frontend	React, Next.js, React Three Fiber, Tailwind CSS, HTML, CSS
Backend & APIs	FastAPI, Flask, Spring Boot, REST API design, JWT authentication
Data Science / ML	Pandas, NumPy, scikit-learn, TensorFlow/Keras, OpenCV, anomaly detection, NLP
Data & Messaging	MySQL, MongoDB, SQLite, MQTT
Tools	Git/GitHub, Docker, Postman, VS Code, Linux
Spoken Languages	French (B2), English (B2)